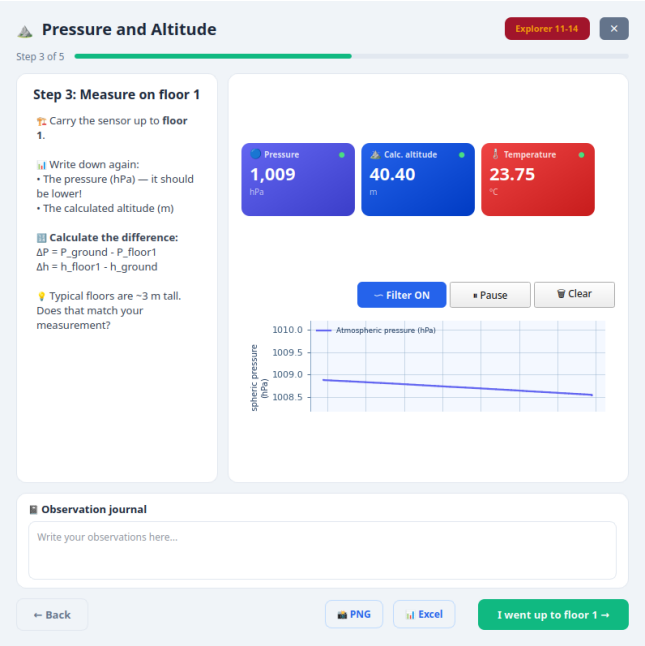
Pressure and Altitude

How to measure the height of your own school with a sensor that only knows how to weigh air

Level: Explorer — ages 11–14 (Romanian school years V–VIII) • **Time in class:** 20 min, two 50-minute lessons or three 50-minute lessons • **Sensors:** BME280 (pressure, temperature, humidity)



The activity screen. The values in the picture are simulated; your own will differ.

Before you start

Who it is for

Explorer — ages 11–14 (Romanian school years V–VIII). This is the Explorer-level entry point into atmospheric pressure: it works with the rule of 0.12 **hPa per metre** (1.2 hPa per 10 m) and with the *difference between floors*, not with logarithms or exponents. The same physics returns at Scientist level, with the full barometric formula and the scale height, in the activity *Physics of the Atmosphere*; this is not a shortened version of that one. The most direct curriculum entry point is **year VII**, where atmospheric pressure is explicit syllabus material.

Assumed knowledge

Calculating a difference, a division and expressing a deviation as a percentage. For the extension, in addition: plotting points on a coordinate system; the formula with a decimal exponent stays an advanced note, for year-X students or the very curious. For support: only "bigger / smaller" and "up / down".

Time in class

Short: 20 min. Full: two 50-minute lessons (1 h 40 min). Project: three 50-minute lessons (2 h 30 min).

Teacher preparation

15 min — walking the stairwell and choosing the landings, measuring one step riser and counting the steps between two floors, finding the school's true altitude, printing the sheets, and getting the building manager's agreement.

Sensor settling time	**30–60 seconds** if the temperature does not change, but **1–2 minutes on every landing** — and if the stairwell is noticeably colder than the classroom, 2–3 minutes. This is the single most common reason for bad data.
Sensors and boxes	BME280. One box per group; **a second box, left on the ground floor, is the single best investment in the whole activity**, because it measures how much the weather changed while the class was climbing.

Three routes

Short	20 min: the written prediction, the rule of 1.2 hPa per 10 m, the ground-floor reference, the climb to the top floor, the return to the ground floor and the conclusion.
Full	Two 50-minute lessons: lesson 1 is the theory, measuring the stairwell with the tape measure, and floor 1; lesson 2 is the full traverse with the return to the ground floor and the height calculation.
Project	Three 50-minute lessons: the two above, plus the analysis lesson — the pressure-against-height graph, the QNH adjustment in **Settings** and, as an advanced note, the full barometric formula on a calculator.

1. Where this fits in the curriculum

Very few school measurements can be checked directly against reality. This one can: students carry a sensor up a staircase, watch a number fall, convert it into metres, and compare the result with a tape measure. The activity fits:

- **Physics** (*Fizică*) — at **year VII**, where pressure, hydrostatic pressure and **atmospheric pressure** are explicit syllabus material, the activity sits exactly on top of the syllabus: pressure is the weight of a column of fluid, divided by area, and here the column is air and it gets shorter as you climb. Archimedes' force, from the same unit, explains why air behaves like a fluid. At **year VI** it connects to mass, volume and density — the density of air, 1.2 kg/m^3, is the only number the derivation below needs.
- **Geography** (*Geografie*) — at **year V**, the Earth's atmosphere: atmospheric pressure, air temperature and how they change with altitude. The link to relief steps and to altitude on maps is an *application* of the measured result, not a separate syllabus entry.
- **Mathematics** (*Matematică*) — calculating a difference, dividing to get a rate (hPa per metre) and expressing a deviation as a percentage at **years V–VI**; plotting points on a coordinate system at **year VII**. The full barometric formula, with its decimal exponent, is only an *advanced note* for very curious students — powers with a rational exponent and logarithms are not studied until **year X**, so it is not a lower-secondary requirement. The Explorer route rests only on the rule $\Delta P \approx 0.12$ hPa per metre.
- **Technology education and practical applications** — the aircraft altimeter, the hiking watch, and the pressure sensor inside a phone that counts the floors you have climbed. They are all the same instrument.
- **Cross-curricular** — the difference between a *reading* and a *result*, the idea that an instrument can be simultaneously poor for the absolute value and excellent for the difference, and the honesty of writing down what the instrument showed, not what it should have shown.

The core skill is **indirect measurement**. The sensor never measures height. It measures pressure, and a simple rule turns that into metres. A student who leaves this lesson knowing that the number on the screen has passed through a model — and that the model has assumptions that can fail — has learned something that transfers to almost any instrument.

2. Time plan

The in-app wizard runs in twenty minutes if you have a staircase right beside the classroom. What turns the activity into a genuine investigation are two things: **measuring the stairwell with a tape measure**, done *before* any sensor reading, which the app cannot ask for, and **returning to the ground floor** at the end, which now has its own row in the app's table, **Ground floor (return)**.

Route	Time	What happens
Short	**20 min**	The written prediction (3), the rule of 1.2 hPa per 10 m (4), the ground-floor reference (4), the climb to the top floor and the reading (6), the return to the ground floor and the conclusion (3). Exactly the app's steps, nothing more.
Full, lesson 1	**50 min**	The written prediction with a number and a unit, and discussing it (8), pressure as the weight of the air column and the rule worked out by calculation (10), **measuring the stairwell with the tape measure** — the riser height and the number of steps (12), the ground-floor reference (8), the climb to floor 1 and the first difference (12).
Full, lesson 2	**50 min**	Traversing the stairwell, stopping on every landing, with **the return to the ground floor** (25), computing Δh with the simple rule and the percentage comparison with the tape measure (12), the discussion of weather drift and of the predictions (13).
Project, lesson 3	50 min	The class's shared graph, pressure against height, and its gradient in hPa per metre (20); the aircraft altimeter, the QNH adjustment in **Settings** and the one-page explanatory text (20); as an advanced note, the full barometric formula on a calculator, for those who can manage it (10).

The full route means **two 50-minute lessons** — an hour and forty minutes — and the project route adds the third, so **three 50-minute lessons**, two and a half hours of contact time. The short route fits comfortably inside a single lesson alongside other content, provided you already have permission to use the stairwell.

The 25 minutes of the traverse are not an optimistic estimate. For a four-storey building they break down like this: about **five minutes at the ground floor** (two to settle, two readings thirty seconds apart, recording), **three and a half minutes at each intermediate floor** (one minute to climb, two to settle, half a minute to record), **five minutes at the top floor**, and **four minutes for the return to the ground floor**. With two intermediate floors this comes to $5 + 3.5 + 3.5 + 5 + 4 = 21$ minutes, leaving four minutes of slack for whatever always happens.

The single most important design decision: the ground-floor box

If you have two boxes, leave one **stationary on the ground floor**, switched on, for the whole traverse. Atmospheric pressure drifts on its own: on an ordinary day by 0.1–0.3 hPa per hour (plus a natural daily swing of about one hPa), and when a front passes by one or two hPa over a school morning — exactly as much as the whole signal from a three-storey climb. Without a control box, students cannot tell a height change from a weather change apart. With one, they subtract the drift and what is left is genuinely theirs.

If you have only one box, do the same thing in time instead: one reading at the ground floor *immediately before* the climb and another *immediately after* the descent, then use the average of the two as the reference. This is the classic surveyor's technique — *closing the loop* — and it is worth naming as such in front of the class, because it gives students the

feeling that they are doing what grown-ups do.

3. Materials

What you need	Per group	Notes
MagicBox with BME280	1	The travelling box. Hold it by the case; fingers on the sensor board warm the temperature channel and shift the pressure reading slightly too.
Second box, control on the ground floor	1 per class	Left switched on, on the ground floor, for the whole traverse, to record the weather drift. **Strongly recommended**; see the teaching tip above.
Laptop with the app	1	Climbs together with the box, so a charged battery matters. It is carried by one named person who does nothing else — nobody carries a laptop and writes at the same time.
USB cable, 2–3 m	1	Lets the box sit on a step or windowsill while the laptop stays on the landing. If you have WiFi in the stairwell, use it — concrete weakens it, so test in advance.
5 m tape measure	1	For the height of a single step riser. **This is the only independent check in the whole activity** and that is why it is not optional.
Clipboard, pencil, the sheet	1 / student	Writing happens standing on a staircase; a pen skips on a sheet held in the hand, a pencil does not.
Watch or stopwatch	1	Every reading carries the exact time, otherwise the drift correction cannot be made.
Sticky tape or chalk	a few	To mark on every landing the exact spot where the box is placed, so the return reading is taken from the same point.
Access to the stairwell	shared	Arranged in advance with the building manager. **Three floors is the practical minimum** for a clear result.
The school's true altitude	1	Look it up in advance on a map and **keep it to yourself** until after the students have read the app's altitude card.
Pocket calculator	1 / group	A simple one is enough for the ordinary route. The full formula, for the extension, needs an x^y key.

Preparation the day before: get permission to use the stairwell and choose a slot *in the middle of a lesson*, not at a changeover — a stairwell during a break is chaos and a hazard. Walk the route yourself and decide which landing each group will use, so that no two groups end up on the same landing. Measure one step riser and count the steps between two floors, so that you know the true height in advance; in an ordinary school a storey is between 3.0 and 3.6 m. Check the forecast: a day with a front passing through produces large pressure drift and is the worst possible

choice, while a settled anticyclonic day is ideal. Finally, switch on a box and leave it undisturbed on your desk for ten minutes: at Explorer level the app shows pressure to one decimal place, and the last digit may flicker by 0.1 hPa; if it jumps by more than that at rest, you will not be able to distinguish a single floor, and it is better to know that a day in advance.

4. Scientific background for the teacher

Where the rule of 1.2 hPa per 10 m comes from — with no integral at all

Pressure at a point is the weight of the air column above that point, divided by the area it presses on. That is all. The rest is year-VI arithmetic.

Take a column of air 10 m tall, with a base of one square metre. Its volume is 10 m^3. The density of air near the ground is $\rho \approx 1.22$ kg/m^3, so the mass of the column is

$$m = 10 \times 1.22 = 12.2 \text{ kg},$$

and its weight is

$$G = mg = 12.2 \times 9.81 \approx 120 \text{ N}.$$

Divided by one square metre, this is a pressure of 120 Pa, that is

$$\boxed{\text{1.2 hPa per 10 m}} \qquad \text{or, equivalently,} \qquad \text{0.12 hPa per metre.}$$

When you climb 10 m, exactly this column is left below you and no longer presses on you. This is not a memorised formula: it is a weighing, and students can do it themselves on paper in three lines.

For scale, the same arithmetic over the whole atmosphere: 1013 hPa means 101 300 Pa, so 101 300/9.81 ≈ 10 300 kg of air on every square metre — **over ten tonnes**. The question "why are we not crushed?" is worth asking out loud; the answer is that air presses equally hard from every side, including from the inside out.

Where the rule breaks down. ρ itself falls with height, so 0.12 hPa per metre is a *local* value. Over a few storeys it is excellent; on a mountain of 3000 m it starts to overestimate, and at the summit of Everest the same ten metres cost only about 0.04 hPa. The full formula, which accounts for this, is the one used at Scientist level.

4.1 Absolute altitude is not the result; the difference is

This distinction decides whether the lesson succeeds, and it must be said *before* the climb, not after.

The app shows a card called **Calc. altitude**. It uses the sea-level pressure set in **Settings** — 1013.25 hPa, the standard value, by default. Today it usually is not that: a strong anticyclone or a deep depression shifts it by 10–30 hPa. Because near the ground one hPa means about 8.3 m, a 10 hPa deviation in the reference shifts the displayed altitude by about 83 m, and on extreme days the error reaches ±100–250 **m**. Students will read "we are at −40 m" or "we are at 310 m" when they know the school is at 85 m, and they will conclude the sensor is broken.

It is not. The **difference** between two altitudes measured a few minutes apart with the *same* box is, however, very reliable, because the unknown reference pressure cancels almost completely. Build the whole lesson around Δh and treat the absolute number as a curiosity to be explained, not a result to be believed.

4.2 How big the signal is, and what limits it

A school storey is between 3.0 and 3.3 m, so the pressure change over one floor is

$$3.0 \times 0.12 = 0.36 \text{ hPa} \quad \text{to} \quad 3.3 \times 0.12 = 0.40 \text{ hPa}.$$

Three floors, about 10 m, give 1.2 hPa; four storeys of 3.2 m each mean 12.8 m and $12.8 \times 0.12 = 1.54$ hPa.

The BME280 resolves changes of about 0.01 hPa and has, when still, its own noise of a few hundredths of a hPa. Converted to metres, 0.03 hPa means $0.03/0.12 = 0.25$ m: the sensor comfortably resolves a single floor.

What you see on screen. At Explorer level the app shows pressure to **one decimal place** (two at Scientist level), so one storey shows up as a drop of 0.3 or 0.4 hPa, and two "identical" readings may differ by one unit in the last digit. The **Calc. altitude** card also has one decimal place, but of a *metre*: 0.1 m is about 0.01 hPa, so on screen the altitude differences are finer than the pressure ones. The pressure chart always spans at least 1 hPa vertically, so that noise does not look like a mountain; a storey shows up on it as a clear step.

So the sensor is not the limit, the weather is. An ordinary pressure tendency is 0.1–0.3 hPa per hour, that is up to 0.1 hPa over the twenty-five minutes of the traverse — almost a metre of spurious height. When a front passes, the drift can reach 0.3 hPa in the same time, that is $0.3/0.12 = 2.5$ m — ten times more than the instrument's own noise. Ask students to compute both contributions and say which one dominates. The answer — that the environment, not the instrument, sets the limit — is a genuinely advanced idea, within reach of a thirteen-year-old right here.

4.3 Why you cannot see the cooling with height

Students expect the top floor to be colder, "because it is higher". The atmosphere's vertical thermal gradient is about 6.5 °C per kilometre, so over 12.8 m it would give $0.0128 \times 6.5 \approx 0.08$ °C — eight hundredths of a degree. The box's thermometer instead shows a difference of one to three degrees between the ground floor and the top floor, tens of times larger, and *in the opposite direction*: it is usually warmer up top, because the building's heated air rises up the stairwell. What the thermometer measures here is the central heating, not the atmosphere. Predicting this failure before the measurement, and then observing it, is a better lesson than pretending it works. The app's conclusion says the same: the atmosphere's cooling of about 6.5 °C per 1000 m is real, but it cannot be seen over a few floors.

5. Lesson flow

The in-app wizard has **seven steps**: the introduction with the rule of 0.12 hPa per metre; a **written prediction**, which the app keeps in the pupil's journal and quotes back in the conclusion; the ground floor; floor 1; the highest point, with the comparison table and the return to the ground floor; the **How sure are we?** step, which asks for two ground-floor readings and reminds pupils to compare differences measured with one box; and the conclusion. Below is what the teacher does around the steps. The times are for the two 50-minute lessons; for the twenty-minute short route, keep points 1, 2, 4, 6, 7 and 9 of the numbered list below (this guide's points, not the wizard's steps), at the times in the time-plan table.

1. **The written prediction, before anything else (8 min).** Before any box is switched on, every student writes three answers: *will the pressure at the top of the school be higher or lower than at the ground floor? by how much, in hPa? and how would you know if you were wrong?* For the second question, ask for a **number with a unit**. Almost every class overestimates massively — you will hear tens of hPa — and confronting that with 1.2 hPa later is one of

the most productive moments of the lesson. Write the tally on the board and leave it there; the two extra minutes are for hearing two or three different justifications. The prediction also goes into the app's prediction box, which will quote it in the conclusion.

2. **What pressure is, by weighing (10 min).** Work through the wizard's first step, then do the weighing from the scientific-background box on the board: a 10 m column, one square metre, 12.2 kg, 120 N, 1.2 hPa. Students now have a *quantitative prediction* to test, which is a far stronger position than "it will go down a bit". The activity is opened from the **Activities** page, with the **Start** button on the activity card.

Technical note — the sensor and the buttons, before the first reading

The sensor. BME280, which gives temperature, humidity and pressure from a single chip. Nothing extra is plugged in. Three cards appear on screen: **Pressure** (in hPa, to one decimal place; two at Scientist level), **Calc. altitude** (in m, to one decimal place) and **Temperature**, with the pressure-over-time chart below them, its time axis in minutes and seconds from the start of the step.

How long to wait. 30–60 seconds if the box has not been moved, but **1–2 minutes on every landing** — and if the stairwell is noticeably colder than the classroom, 2–3 minutes. The app times the wait itself: on every measurement step a yellow strip counts down **90 seconds** ("Let the box settle: … s more") and turns green when the time is up; the **I moved the box** button next to it starts the count again after every move. The reason is not that the chip "warms up": the pressure is temperature-compensated inside the sensor. The reading keeps drifting because it is *smoothed* — the sensor's filter and the app's averaging need a few tens of seconds to reach the new value — and in a stairwell, draughts and opening doors move the pressure by a few hundredths of a hPa.

The buttons you will use. The activity moves forward with the green button at the bottom right, and **its label changes at every step**, because it echoes what you have just finished: **Got it!**, **My prediction is written**, **Ground floor noted**, **I went up to floor 1**, **I measured at the highest point**, **Continue**, and at the last step **Finish**. The **Back** button, on the left, goes back one step. On the measurement steps, the **Capture** button copies the reading on screen into the current row of the comparison table. Above the chart sit **Smooth: on / Smooth: off**, **Pause** — which freezes the chart — and **Clear**.

Exporting the data. Two buttons appear on the measurement steps: **PNG** saves the charts as an image, and **Excel** saves the step's *raw* (unfiltered) readings to an `.xlsx` file, with an `elapsed_s` column — the seconds since the start of the step. The step with the comparison table has no chart, so export the chart from the measurement steps before moving on. The pupil's prediction and journal notes are saved automatically, to disk (in the `logs` folder, a `…_journal.txt` file) and as a **Journal** sheet in the activity's Excel file. The box's complete recording is taken from the **Data & Export** page; exports default to `.xlsx`, and a CSV saved while the app is in Romanian uses ";" and a decimal comma, so that it opens correctly in a Romanian Excel. Save using the pattern `class_date_activity` — for example `7A_2026-09-14_presiune_altitudine`. Note on the sheet the building, the starting floor, the time of every reading and the measured height of one storey.

3. **Measuring the stairwell with the tape measure, before the sensor (12 min).** Groups measure **one step riser**, count the steps between two floors, and multiply. This is the independent truth against which the sensor will be judged, and doing it *first* prevents the temptation to fit the answer afterwards. Write on the board the value the whole class adopts; if two groups get different numbers, discuss why before choosing one.

4. **The ground-floor reference (8 min).** The travelling box is placed on a spot marked with tape, on the ground floor, and you wait the 1–2 minutes. Students record **four** things: the pressure, the altitude on the card, the temperature and the **exact time**, then press **Capture** so that the **Ground floor** row of the app's table fills in. Insist on the time; without it, the drift

correction is impossible. If you have a second box, this is also when you switch it on and leave it there. Tell them plainly: **a measurement without a reference is not a measurement.**

Technical note — right at the step where the altitude card appears

The Calc. altitude card depends on an assumption, not only on where you are. The app computes altitude from pressure with the barometric formula, using the sea-level pressure from **Settings** — 1013.25 hPa by default. On a real day this may be 995 or 1030 hPa, and since near the ground one hPa means about 8.3 m, a 10 hPa deviation shifts the displayed number by about 83 m, and on extreme days the error reaches ±100–250 m. A group can therefore get "we are at −40 m". The sensor is not broken and the students have not made a calculation mistake.

The rule to use for the rest of the activity. Absolute altitude is not the truth here; what is being measured are the *differences* between floors. The difference between the ground floor and the top floor hardly depends on the assumed reference, because the same reference pressure cancels out. This is why you never compare one box's absolute value with another's: two BME280 units can differ by 1–2 hPa in absolute terms while still tracking the same changes flawlessly.

The reference can be adjusted, and it is worth showing. On the **Settings** page, the **Altitude Calibration** section has a **Sea-level pressure (QNH)** field, starting from 1013.25 hPa; the value you choose is kept when the app restarts. If you change it until the card shows the school's true altitude, the number you land on is today's actual QNH. Example: a school at 85 m measuring 1008.0 hPa needs about 1018 hPa, which matches the pocket rule $1008 + 0.12 \times 85 = 1018.2$ hPa. This value is then compared with the national forecast — the only external check available in the classroom. Remember to set it to the day's value next time: QNH changes from one day to the next.

The comparison table. At the highest-point step the app shows a table with the columns **Location**, **Pressure (hPa)**, **Altitude (m)** and **ΔAlt (m)** and **five rows**: **Ground floor**, **Floor 1**, **Floor 2**, **Floor 3** and **Ground floor (return)**. The Ground floor and Floor 1 rows are already filled by the **Capture** button of the earlier steps. Here the **Row to fill:** list chooses the row, and **Capture** fills it with the reading on screen and moves on to the next empty row by itself; capturing again on a filled row replaces it. Press it only after the waiting strip has turned green. The app's table does not replace the sheet: the time of every reading, the second reading at the ends and the tape-measure height are written on paper, in pencil.

5. **The first climb: floor 1 (12 min, lesson 1).** The group climbs a single storey and repeats exactly the same routine: places the box on the marked spot, waits, records four numbers. It then computes on the spot the pressure difference from the ground floor and compares it with what it had predicted. A single storey gives 0.36–0.40 hPa, and the most common reaction is disappointment. Reframe it: the number is small, but the instrument found it without any doubt, and that is exactly what is worth celebrating.

6. **The full traverse (25 min, lesson 2).** The group climbs floor by floor, with the same routine at every landing. Stagger the groups by one floor, so that no two are ever on the same landing — a crowd of warm bodies on a small landing is itself a measurable disturbance. The box is not held in a warm hand between readings: it is carried in a bag or on the clipboard.

   Two readings, but only at the ends. At the ground floor and at the top floor, take **two readings**, about thirty seconds apart, *after* the number has stopped drifting — you do not wait another two minutes for each one. At the levels in between, a single reading after settling is enough: they give the shape, not the spread. The sheet has its rows printed exactly for this pattern.

7. **The return to the ground floor — the step that is not optional (included in the**

25 min). Everyone comes back down to the spot marked at the start and takes a third reading. The difference between the first and the last ground-floor reading — the **Ground floor (return)** row in the app's table — is **the weather drift, measured, not assumed**. Students almost always find it non-zero, and the discussion that follows — *"so which of our numbers do we actually believe?"* — is the heart of the lesson. If the drift exceeds the difference between two neighbouring floors, compare only the extremes, not adjacent floors.

8. **Computing and comparing (12 min, lesson 2).** Each group computes ΔP between the ground floor (the mean of the two readings) and the top floor, divides it by 0.12 to get metres, and compares the result with the tape-measure height, expressing the deviation as a percentage. Each group then states *one* reason for its own deviation.

9. **Conclusion and the connection outward (8 min).** Go back to the tally of predictions on the board and compare it with 1.2 hPa. Close with the app's conclusion step, which quotes the prediction written at the start and whose green button reads **Finish**, and with the connections outward: an aircraft's altimeter is a barometer, which is why the pilot is given the local pressure by air traffic control before landing and why every aircraft in a region flies on the same setting; a barometric altimeter works in a tunnel, in a forest and inside a building, where satellite signal does not reach — which is why modern phones carry one, to count the floors you have climbed.

10. **The graph and the full formula (lesson 3 — project).** Plot pressure against the true height and extract the gradient in hPa per metre; over a few storeys the relationship is indistinguishable from a straight line. As an advanced note — the decimal exponent is year-X material — very curious students can check on a calculator that the full barometric formula gives the same Δh as the simple rule.

The investigation, in one table

The question. By how much does air pressure fall for every metre climbed, and how tall is our building, measured with air?
The hypothesis. Written on the sheet **before** any measurement, with a **number and a unit** for the drop over the whole climb, and not changed afterwards.
What I change. One thing only: the **level** the box is on, expressed in metres above the ground floor and obtained from the tape measure, not from the sensor.
What I measure. Pressure, in hPa, to one decimal place, as the app shows it at Explorer level (two at Scientist level). The altitude on the card — which, with its decimal place of a metre, shows finer differences than the pressure — the temperature and the **exact time** of every reading are also recorded.
What I keep the same. The same stairwell, the landing doors closed, **the same box** from start to finish, the same marked spot on every landing, the same waiting time, and the same person reading the screen.
How many repeats. Two readings at the **ground floor** and two at the **top floor**, thirty seconds apart, *after* the number has stopped drifting. At the intermediate levels, a single reading after settling. Plus the **mandatory return to the ground floor** at the end.
What makes a result valid — and what makes it an artefact. Compare the spread (maximum minus minimum) of the two ground-floor readings with the pressure difference between the ground floor and the top floor. **If the spread at a single level is bigger than the difference between levels, the correct conclusion is that the difference is not detectable with your setup — not that it does not exist.** These are two different statements, and the distinction between them is exactly what is being taught here. The remedy is not to hunt for nicer numbers in the table, but to increase the signal (climb more floors, compare the extremes rather than neighbouring floors) or to reduce the noise (wait longer, keep the landing doors closed).

The closing rule. If the reading on returning to the ground floor differs from the first one by more than the spread of the two initial readings, **the weather changed during the measurement** and the raw result cannot be trusted. Subtract the drift — proportionally to the time elapsed — or repeat the traverse on another day. This is not pedantry: on a day with a front, a drift of 0.3 hPa means 2.5 m of spurious height, almost a whole storey.

6. Analysis after the activity

6.1 What the data should look like

The values below assume a school building with 3.2 m storeys, on a settled day.

The situation	Typical reading	Interpretation
Pressure at the ground floor	960–1030 hPa	Depends both on the school's altitude and on the weather. Whatever it shows today becomes the reference.
The drop over a single storey	0.36–0.40 hPa	For 3.0–3.3 m. Small, but comfortably above the noise.
The drop over three storeys (about 10 m)	about 1.2 hPa	Exactly the rule obtained by weighing the air column.
The drop over four storeys (12.8 m)	about 1.5 hPa	$12.8 \times 0.12 = 1.54$ hPa.
The **Calc. altitude** card, step per floor	3–4 m	The difference is correct even though the absolute value is not.
The **Calc. altitude** card, absolute value	can be 100–250 m off	Expected, not an error: sea-level pressure is not 1013.25 hPa today. Corrected with the QNH in **Settings**.
The return to the ground floor	within 0.1 hPa on a settled day; 0.3 hPa or more with a front	The difference is the weather drift; subtract it.
The instrument's own noise, box still	a few hundredths of a hPa; on screen the last decimal flickers by 0.1	Under 0.3 m converted to height. The sensor is not the limit.
The gradient over the whole climb	-0.11 to -0.13 hPa/m	Theory gives -0.120. Agreement within ten per cent is a good classroom result.
Temperature, ground floor versus top floor	warmer up top by 1–3 °C	The building's heating, not the atmosphere: the atmospheric gradient would give only 0.08 °C over the whole building.

Because a single storey gives only about 0.4 hPa, a group that compares only two neighbouring floors on a windy day may get an ambiguous or even reversed result. Always climb as far as the building allows, and **compare the extremes, not neighbours**, when the signal is marginal.

6.2 From pressure to metres, step by step

1. Enter into the table, for every level, the time, the pressure in hPa to one decimal place, the altitude on the card and the temperature.
2. **Weather drift** = the second ground-floor reading − the first. If it is small compared with the difference between levels, carry on using the mean of the two ground-floor readings as the reference.
3. ΔP = the ground-floor reference pressure − the pressure at the top floor. Worked example, with typical figures: $1008.0 - 1006.5 = 1.5$ hPa.
4. The height, from the simple rule: $\Delta h = \Delta P / 0.12$, that is $1.5/0.12 = \mathbf{12.5\ m}$. For students who find dividing by 0.12 strange, the same operation can also be written as "for every 1.2 hPa, 10 m": $1.5/1.2 \times 10 = 12.5$ m. With a single decimal place, ΔP is only certain to about ± 0.1 hPa, that is ± 1 m; the difference between the altitudes on the card gives the same height, more finely.
5. Compare with the tape measure. Four storeys of 3.20 m mean $4 \times 3.20 = 12.80$ m, so the deviation is $12.80 - 12.5 = 0.30$ m, that is $0.30/12.80 \times 100 = 2.3\%$. Ask each group for one reason for its own deviation.
6. **Advanced note, on a calculator** (the decimal exponent is year-X material): check that the full barometric formula,

$$h = 44330 \left[1 - \left(\frac{P}{1013.25} \right)^{0.1903} \right] \text{ m},$$

   gives the same result. With $P = 1008.0$ hPa this gives $h = 43.8$ m, and with $P = 1006.5$ hPa it gives $h = 56.4$ m; the difference, $56.4 - 43.8 = 12.6$ m, deviates from the simple rule by under one per cent. Both absolute numbers are wrong if the school is not at that altitude — their *difference* is not.
7. **Extension:** plot pressure (vertical) against true height (horizontal), scaling the pressure axis *only* over the range of the data — an axis starting at zero flattens the whole effect. The axes are prepared on the sheet for exactly this reason. The gradient of the line is the class's own value for -0.12 hPa/m.

6.3 Questions to ask afterwards

1. The app said our school is at a certain altitude, but the map says something else, and the sensor is working perfectly. Explain the mismatch, and explain why the *difference* between two floors nonetheless came out correct.
2. The two ground-floor readings, taken twenty minutes apart at the same marked spot, were not identical. What changed? How would you correct the rest of your data for it?
3. What limited you more — the sensor's own noise, or the weather? Support your answer with numbers from your own table.
4. Estimate how much the pressure would fall if you climbed a 300 m hill. Would the rule of 1.2 hPa per 10 m still be a good approximation at 3000 m? Use the fact that air is compressible.
5. The top floor came out *warmer* than the ground floor, even though it is higher up. Explain why, and say what the atmosphere alone should have shown over this height.
6. An aircraft flies at a constant indicated altitude and passes from a high-pressure region into a low-pressure region, without the pilot changing anything. Does its true height above the ground go up, go down, or stay the same? Make a sketch.

6.4 Ideas for assessment along the way

- **Low stakes:** the student completes a partly filled table, computes ΔP between the ground floor and the top floor, converts it into metres with the rule of 1.2 hPa per 10 m, and says whether the answer is bigger or smaller than the tape-measure value. No percentage is required.
- **Standard:** a one-page write-up with the prediction, the method (including the return to the ground floor), the table of stations, the calculation of Δh, the percentage deviation from the tape measure, and *one* named source of error, with its size estimated in metres.
- **Extension:** the student plots pressure against height, extracts the gradient and compares it with -0.12 hPa/m; adjusts the **Sea-level pressure (QNH)** field in **Settings** until the altitude card shows the school's true altitude, and compares the value found with the day's national forecast; or, as an advanced note, checks the full barometric formula on a calculator.

Common pitfalls and misconceptions

"The altimeter says we are at −40 m, so it must be broken." The single most common reaction in the whole activity. Absolute altitude assumes that today's sea-level pressure is 1013.25 hPa. Clear this up *before* the climb, not after.
Expecting a big number. Students predict tens of hPa for three storeys and are disappointed by 1.2. Set the expectation quantitatively in advance, by weighing the air column together, so the small number is a confirmation, not a letdown.
Reading before the sensor has settled. A box moved to another landing needs a minute or two, because the reading is smoothed and reaches the new value gradually. A reading taken immediately — or a **Capture** pressed immediately — is still partly the reading of the previous spot.
Forgetting the time. Without the exact time, the drift correction cannot be applied, and the best part of the analysis is lost.
Ignoring the weather drift. A twenty-minute traverse during a passing front can produce a pressure change bigger than the whole climb, and in the wrong direction. The return to the ground floor is not optional; it is the difference between data and noise.
Comparing only two neighbouring floors. A single storey is close to the noise floor on a bad day. Always also compare the extremes.
Confusing pressure with temperature. Students expect the top to be colder, "because it is higher". In a heated building it is usually warmer. Predict this, then discuss why the atmosphere's cooling — 0.08 °C over the whole building — is undetectable here.
Opening an outside door in the wind. It shifts the pressure inside the building by a few tenths of a hPa, the equivalent of a whole storey. If it happens, note it on the sheet instead of discarding the reading.
Comparing boxes with each other. Two BME280 units can differ by 1–2 hPa in absolute terms. Compare *differences*, never absolute values, between boxes.

Safety and classroom management

The activity moves children through a stairwell carrying fragile equipment, so the risk is higher than for an ordinary classroom practical. Treat the stairwell as a field site.
Nobody reads a screen while walking. This is the central rule of the lesson and is stated plainly before leaving the room, then repeated at the first slip. The box is **carried**, it is read **standing still, on a landing**. Whoever carries the laptop watches where they step and reads nothing while moving. Nobody writes on the sheet while going down the stairs.
Stairs and traffic. Never run the traverse at a lesson changeover. Groups keep to the wall side, use the handrail, and stop **only on landings**, never on the flight itself, where a stationary group is a serious obstruction. Nobody runs.
Supervision. Ideally, one adult per floor in use. As a minimum: one adult on the stairwell

and one who stays with the control box and the students left in the classroom, with at most four students in a travelling group, so that a group is never split. Groups must not be sent to floors that are out of sight and out of earshot of any adult.
Roof and high-level work. Only use the roof if it is a designed space, with a parapet and routine access, and only with an adult present at all times and the explicit agreement of the building manager. Otherwise stop at the highest indoor landing — the physics is identical.
Cables. A USB cable between the laptop and the box on a staircase is a real trip hazard. Use the shortest workable arrangement, keep the cable off the steps, or work over WiFi.
Equipment. A box dropped on a stone staircase will break. Carry it in a bag or a box between stations, never loose in a hand on the stairs.
Fire doors and access. Agree the route with the building manager in advance and do not prop fire doors open.

Adaptations for pupils with special educational needs

The no-division route: only "bigger / smaller". This is the support route and it reaches the same physical conclusion. The student calculates nothing: they only fill in the pressure column of the stations table — whose rows are printed in advance, with the levels already written in — and say out loud, at every landing, whether the value is *bigger* or *smaller* than at the ground floor, while a classmate writes it down. The only decision they write themselves is the *detectable / not detectable* tick in the closing block, and that is exactly the scientific conclusion of the lesson. For extra support, fill in the ground-floor row yourself *before* copying the sheet: the student continues a pattern rather than inventing one.
The conversion, already worked out. Dividing by 0.12 is the real obstacle, not the physics. Print on the sheet the conversion table already computed — 0.4 hPa → 3 m, 0.8 hPa → 7 m, 1.2 hPa → 10 m, 1.6 hPa → 13 m — and let the student read from it instead of dividing. They reach the same scientific result by a different route. Also give them the tape-measure height, so their task is only the comparison.
The settled route, for students with motor difficulties. The whole building is not needed for there to be a result. **Two levels are enough**: the pressure in the classroom, the pressure one floor up, and the return. That is three readings, a single round trip, and the same physical conclusion, only with a smaller signal — allow three minutes instead of two at each stop in that case.
Three real roles, all held from a seat. A student who does not climb is not a student who does not take part. The roles rotate at every level:

- **control-box monitor** — stays on the ground floor with the second box, records the pressure minute by minute and, at the end, tells the class how much the weather changed; this is the person who decides whether the traverse's result can be trusted, so it is the role with the greatest scientific weight in the class;
- **timekeeper** — keeps the clock, announces when the two minutes of settling have passed, and has the right to reject any reading taken too early;
- **data recorder** — keeps the group's table, checks on the spot whether the two readings from the same level are close together, and announces if one needs repeating.

Oral answers instead of writing. The hypothesis, the explanation of the altitude mismatch and the closing sentence can be given **out loud**, with the teacher or a classmate writing them on the sheet. For the sentence frames, the student can complete them orally and only sign the row. What is assessed is the reasoning, not the speed of writing.
Language and cognitive support. Write on the board, at the start, the four words of the lesson — pressure, altitude, difference, drift — and use them unchanged all lesson. Repeat the same sentence every time: *the difference is the result, the absolute altitude is not.* For students sensitive to crowding, the landings can be visited quietly, with small staggered

groups, while the rest of the class works on the conversion table.

If things do not go to plan

You do not have access to a stairwell. Measure the difference between the floor and a tabletop, or between the floor and a cupboard: about a metre, so about 0.12 hPa. The signal is small, so extend the waiting time to four or five minutes at each position and take five readings instead of two. If the spread stays bigger than the difference, that is the honest result of the lesson — the difference is not detectable with this setup — and it is a better lesson in method than a nice number obtained by guessing.

You have one box and thirty students. Do the traverse once, with the whole class, as a single movement. At every landing a different student reads the screen out loud and another confirms that the number has stopped moving; everyone records in their own table. You get a single data set, but the whole class analyses it, and the reading role passes round everyone. The ground-floor reference is then taken before and after, and their mean is the reference.

You have twenty minutes, not fifty. This is exactly the short route. Keep exactly four moments: the written prediction with a number, the ground-floor reading, the top-floor reading, and the return to the ground floor. Drop the intermediate floors, the graph and the full formula — **but not the return**, because without it you cannot tell whether the result belongs to the building or to the weather.

You have no projector. It does not matter much here: the activity happens on the stairs anyway, around a single laptop. Name, at every landing, a **screen reader** who says the four numbers out loud, and a **witness** who confirms that the reading has stopped drifting. The rest of the class records by dictation.

The sensor will not settle. This is normal in the first minute and a half. If the number still drifts after three minutes, the cause is almost certainly a draught in the stairwell: close the landing doors and wait another two minutes, with the box set on a surface, not held in a palm. Make this the topic: why does the number not jump straight to the new value? Because the reading is smoothed, so that noise does not make it dance, and any smoothing means a delay — a trade-off every measuring instrument makes.

The box will not connect. Do not stop the lesson. Measuring the stairwell with the tape measure and the written prediction can still be done, without a sensor. Then use the "What the data should look like" table from this guide as a printed data set: students receive the values row by row, compute Δh, compare it with the tape measure, and discuss the drift. Repeat the measurement on another day, after checking the cable, the port selected in the app and, if you are on WiFi, the network.

7. Further resources

Stable, well-established starting points. Search within each site for the topic named — these are root addresses, not deep links, so they will not rot over time.

- **National Meteorological Administration (ANM)** — `https://www.meteoromania.ro` — search for atmospheric pressure and the surface pressure chart for the day of the lesson; it shows the weather system that produced the drift in the class's data, and it is where you get the value to compare with the QNH set in **Settings**.
- **NOAA** — `https://www.noaa.gov` — search for "standard atmosphere" and the difference between station pressure and pressure reduced to sea level; the clearest explanation of why the app's absolute altitude is offset.
- **PhET Interactive Simulations** — `https://phet.colorado.edu` — search for the simulations on fluid pressure and gas properties; they make the idea "pressure is the weight of the

column above you" visible before anyone climbs any stairs.

- **Institute of Physics** — `https://www.iop.org` — search for practical resources on pressure for ages 11–14; teaching sequences and cheap demonstrations on fluid pressure, to run alongside this measurement.
- **STEM Learning** — `https://www.stem.org.uk` — search for atmospheric pressure and data logging for ages 11–14; classroom-tested sheets and risk-assessment templates for off-site activities.
- **European Space Agency (ESA)** — `https://www.esa.int` — search for the atmosphere and Earth observation, for the jump from a stairwell to the satellites that measure the profile of the whole atmosphere.

8. Assessment

The exit ticket, identical to the one on the student sheet: *one thing that surprised you today, and one thing you would do differently if you repeated the measurement.* It is collected in the last two minutes of the lesson and says more about the class's understanding than any mark.

For a more formal judgement, use the **shared three-strand rubric** — *measurement practice*, *data interpretation* and *scientific reasoning.* At Explorer level, the three strands look like this:

- **Measurement practice** — the student waits for the reading to settle before recording, notes the time of every reading, measures the height with the tape measure instead of guessing it, and makes the return to the ground floor without being reminded.
- **Data interpretation** — the student computes ΔP and converts it into metres with the correct unit, expresses the deviation from the tape measure as a percentage, and distinguishes "the difference is not detectable" from "the difference does not exist".
- **Scientific reasoning** — the student explains why the absolute altitude is wrong while the difference is good, names one source of error and says which way it pushes the result, and does not confuse the building's heating with the atmosphere's cooling.

The rubric itself, with descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

9. The student sheet

The sheet the students fill in is a separate document: `docs/en/student_sheets/pressure_and_altitude.pdf`. It has two pages — the prediction, measuring the stairwell with the tape measure and the stations table on the first; the closing check, the calculation, the graph and the conclusion on the second — so it prints double-sided, one sheet of paper per student. It is built on the three layers — what I see on the screen, what I do on my own, the conclusion I write — and asks for the prediction **before** any measurement.

Three things worth knowing before printing it. **The "Ground floor (return)" row is printed as a mandatory row**, not an optional one, and immediately below the table sits the closing block, where the weather drift is compared with the difference between levels — this is where the student decides for themselves whether they are allowed to believe the result. **The table has its rows already labelled**, with the time column beside the pressure column, and the graph's axes are already drawn, with the warning that the pressure axis does not start at zero. **The hPa → metres conversion table is printed on the sheet**, precisely so that dividing by 0.12 is not the obstacle that stops a student from reaching the physics. The tables are not repeated here precisely so that the two cannot drift apart: there is a single printable version, and it is the sheet.